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How Much Is This House Worth — and How Sure Are We?

House Price Prediction for a Real Estate Listing Platform

IT5315E Business Analytics · Group Project · Final Report

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Live valuation tool: house-price-analytics-tjpb7ntxsd6kzbapp7bymmv.streamlit.app

Source code: github.com/ducnt3/house-price-analytics

1. The business problem

A listing platform loses both ways when prices are wrong: an inflated asking price scares serious buyers away, while an underpriced home leaves the seller's money on the table. Agents need a *defensible* estimate long before a formal appraisal — and, just as importantly, they need to know **how much to trust it**. An estimate of $\$180\text{k} \pm \$18\text{k}$ supports a confident listing; $\$180\text{k} \pm \$70\text{k}$ says "get an appraisal first." Our deliverable is therefore not a number but a number **with a calibrated confidence range**, deployed as a working tool.

Error tolerance. Mispricing by roughly 10% stays within the margin agents negotiate; a systematic bias beyond about 5% at portfolio level distorts the platform's market statistics and must trigger model retraining.

2. The data — and why it can be trusted

Base: the Kaggle Ames, Iowa dataset — 1,460 home sales with 80 structural attributes. **Extension (required synthetic component):** 12 contextual fields generated in Python, each documented in the data dictionary with its type, unit, valid range, and the business assumption behind it.

2.1 A real time axis

Rather than inventing listing dates, we anchored them to the sale dates already inside the Ames data (January 2006 – July 2010). No row contradicts itself, and the window spans a true market story — the financial crisis. A synthetic macro layer (an interest-rate series and a market price index with a boom, a –10% crisis, and a +9.5% rebound in 2010) re-expresses each sale price under that trend:

$$\text{sale\_price}_i \;=\; \text{SalePrice}_i \times \frac{I(m_i)}{\bar{I}} \times \varepsilon_i, \qquad \varepsilon_i \sim \text{LogNormal}(0,\, 0.02)$$

where $I(m_i)$ is the index level in listing month m_i and $\bar{I}$ its window average. The original Kaggle price is preserved in a separate column for auditability. All 175 sales of 2010 are held out as the "incoming stream" for monitoring.

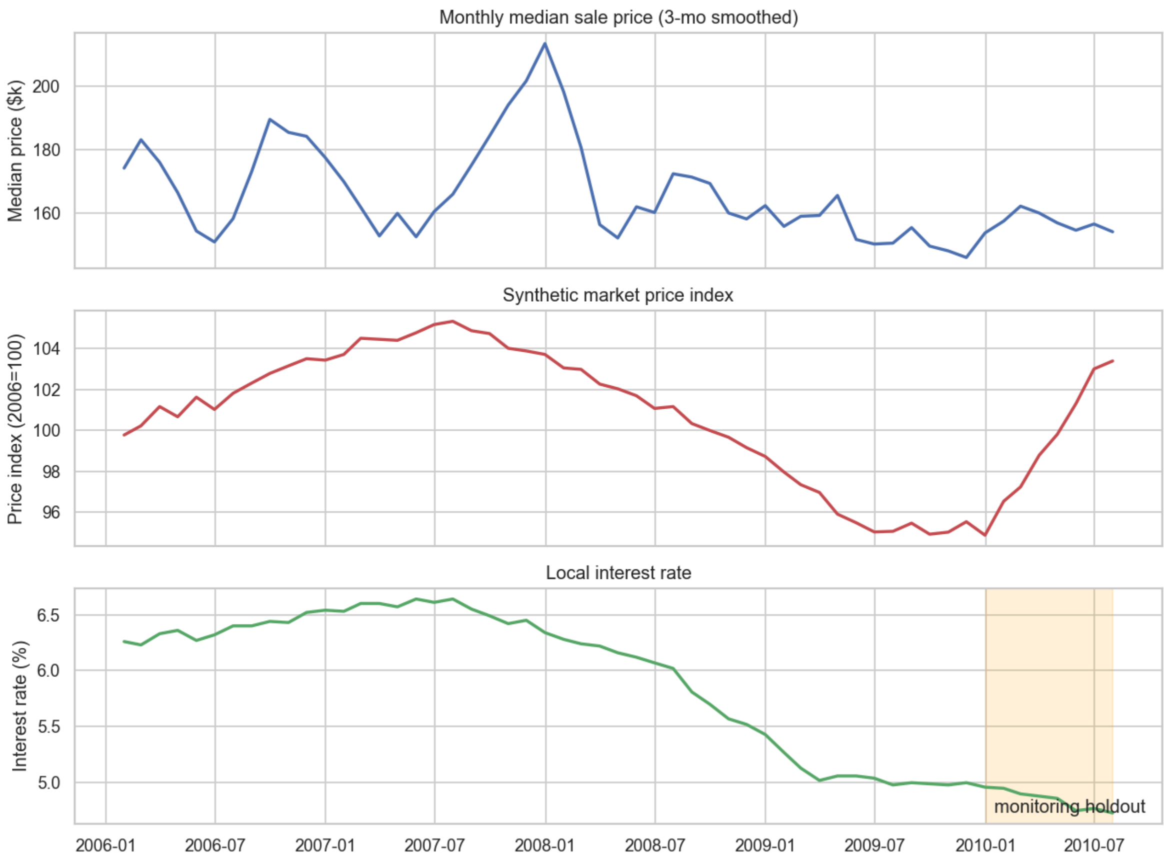

Figure 1 — Median price, synthetic market index, and interest rate over 2006–2010; the shaded band is the monitoring holdout.

2.2 Behavioral realism, generated conditionally

- Amenity distances are a neighborhood property: pricier neighborhoods sit closer to schools (realized correlation –0.60 with price); transit access is deliberately unpriced (r ≈ 0.1) — bus lines follow arterial roads, not wealth.
- Renovation costs are consistent with the dataset's real remodel-year column; days-on-market lengthens when the market index falls, as it did in 2008–09.

2.3 Documented imperfections

Real listing pipelines produce dirty data, so five seeded, documented problems (missing values, free-text label variants, sentinel 999s, metre/km unit errors, duplicate listings) were injected — only into synthetic fields and only before 2010, so the monitoring stream is never confused with dirt. Every claim in this section is enforced by automated tests, and generation reproduces byte-for-byte from a fixed seed.

3. What drives price (EDA)

- **Quality and size dominate.** Overall quality correlates 0.79 with price (median \$85k at quality 3 → \$346k at quality 9); living area 0.71.
- **Location is a 3.5× lever.** Neighborhood medians span \$90k (MeadowV) to \$313k (NridgHt), and the amenity fields explain much of the premium: price falls steadily with school distance while transit distance is flat.
- **The market moved** — up ~5% to mid-2007, down ~10% through 2009, rebounding in 2010. A price model in this market ages; monitoring is not optional.

Sale price is right-skewed (skewness 1.93, falling to 0.12 after the transform), so every model predicts the log target

$$y = \log(1 + \mathrm{SalePrice}).$$

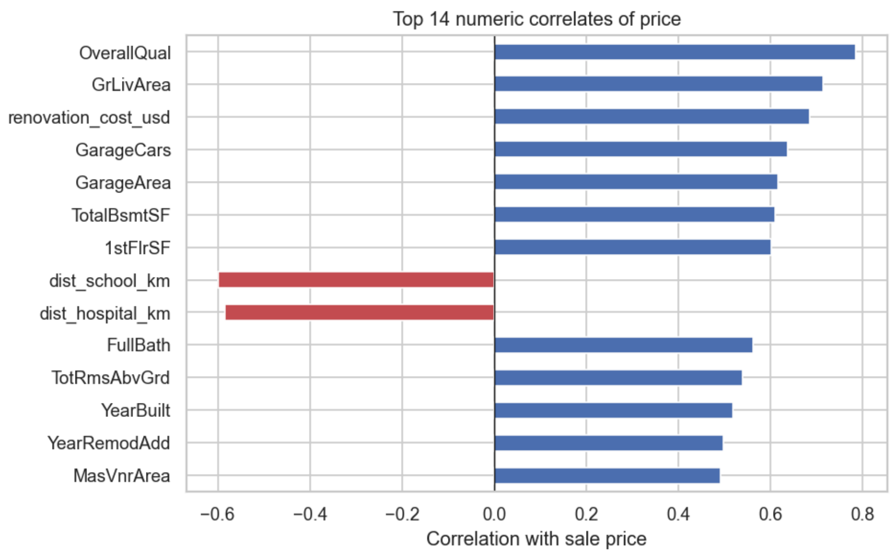

Figure 2 — Top correlates of sale price. Synthetic amenity fields behave as designed.

4. Cleaning, with before/after evidence

Problem (diagnosed cause)	Before	After
Duplicate listings (re-posted homes) — keep the earliest posting	12	0
Free-text renovation labels (Y / yes / NO / …) → canonical Yes/No	51	0
Days-on-market sentinel 999 → listing-quarter median impute	5	0
Transit distance entered in metres → converted to km	4	0
Missing school distance → neighborhood median (a neighborhood property)	26	0
Kaggle "no such feature" NAs → explicit category	7,480 cells	0
LotFrontage → neighborhood median (subdivision plans set frontage)	259	0
Partial-sale outliers (Ids 524, 1299; huge area, far-below-trend price)	2 rows	dropped

1,472 listings in → 1,458 out. Every fix targets a diagnosed cause, and the cleaning script prints this table on every run.

5. Features and multicollinearity

Leakage discipline. Only information available at valuation time enters the model. Excluded: days-on-market (an outcome of the listing, unknown on pricing day) and the market price index (published with a lag — using the current month's value would be look-ahead leakage). Included: the interest rate, which is public when a home is listed.

Derived features: property age, years since remodel, combined bathrooms, renovation flag and cost, amenity-proximity score. We keep one variable per concept (garage cars, not garage area — they correlate 0.88) and verify with variance inflation factors:

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is from regressing feature j on all others. Maximum VIF across the 16 numeric features is **3.9**, far below the conventional threshold of 10 — the linear baselines are stable.

6. The model — and its honesty about uncertainty

6.1 Comparison under 5-fold cross-validation

All models predict log price; metrics are reported on the dollar scale:

$$\mathrm{RMSE} = \sqrt{\tfrac{1}{n}\textstyle\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \qquad \mathrm{MAE} = \tfrac{1}{n}\textstyle\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$\mathrm{MAPE} = \tfrac{100\%}{n}\textstyle\sum_{i=1}^{n}\left|\tfrac{y_i - \hat{y}_i}{y_i}\right| \qquad R^2 = 1 - \tfrac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

Model	RMSE	MAE	MAPE	R^2
Linear Regression	$23,409	$15,806	9.11%	0.914
Ridge (best point model)	**$23,195**	**$15,573**	**8.99%**	**0.916**
Lasso	$23,324	$15,670	9.04%	0.915
Random Forest	$29,509	$18,469	10.43%	0.863
LightGBM (deployed, quantile)	$27,441	$17,334	9.78%	0.882

Ridge wins on point accuracy — log price is close to linear in the drivers, and 1,283 training rows favor low-variance models. **We nevertheless deploy the quantile LightGBM** (its median forecast as the point estimate): the business question demands a per-home confidence range, quantile regression provides feature-dependent intervals natively, and one coherent model beats stitching two model families together. The cost (9.78% vs 8.99% MAPE) is immaterial to the tool's purpose; the comparison is reported, not hidden.

6.2 Calibrated confidence intervals

Three LightGBM models are trained on the pinball (quantile) loss

$$\mathcal{L}_\alpha(y,\hat{y}) \;=\; \max\big(\alpha\,(y-\hat{y}),\;(\alpha-1)\,(y-\hat{y})\big), \qquad \alpha \in \{0.10,\,0.50,\,0.90\},$$

giving a p10–p90 band that *claims* 80% coverage. On a 20% calibration split never seen in training, the raw band actually contained only **59%** of true prices — naive quantile intervals lie. Conformalized quantile regression (CQR) measures the shortfall through the conformity score

$$E_i \;=\; \max\big(\hat{q}_{0.10}(x_i) - y_i,\;\; y_i - \hat{q}_{0.90}(x_i)\big)$$

and widens the band by the empirical $(1-\alpha)$-quantile of these scores:

$$\hat{C}(x) \;=\; \Big[\hat{q}_{0.10}(x) - Q_{1-\alpha}(E),\;\; \hat{q}_{0.90}(x) + Q_{1-\alpha}(E)\Big], \qquad 1-\alpha = 0.80.$$

> **Verified coverage after calibration: 80.5%** (target 80%). Interval width behaves as a buyer would expect: ≈$35k for a typical home, ≈$78k for high-value homes — and the UI translates width into advice (a wide range means "consider a professional appraisal").

Residual analysis shows no systematic bias across price segments except the cheapest decile, which is over-predicted — noted as a limitation.

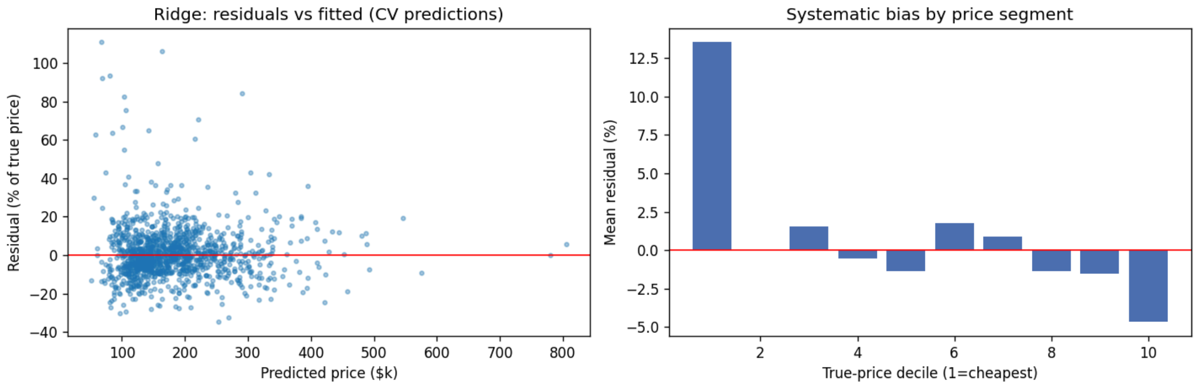

Figure 3 — Cross-validated residuals vs fitted values and mean bias by price decile.

7. The working product

- **Valuation tool** (live link on the cover): a Streamlit app returning the estimate, the calibrated 80% range, and a plain-language confidence note. The model runs in-process — the demo cannot be taken down by a second service failing.
- **Prediction API:** a FastAPI service exposing `POST /predict` with OpenAPI documentation; omitted fields fall back to training-median defaults and the response lists which. Both surfaces share one serving layer, so they can never disagree.
- Deployment survived a real vendor shock: the first target (Hugging Face Spaces) paywalled compute mid-project; because the pipeline was proven in week one with a placeholder model, switching to Streamlit Community Cloud cost hours, not weeks.

8. Monitoring: the model watches itself age

The 175 sales of 2010 were replayed month by month through Evidently, evaluating a rolling 3-month window against the 2006–2009 training baseline. Four retraining triggers were defined *before* looking at the stream:

Trigger	Definition	2010 outcome
T1 — Data drift	>30% of features drift (Evidently; windows of ≥40 rows)	Fires Mar–Jul: interest-rate regime + post-crisis sales-mix shift
T2 — Performance	Rolling RMSE > 1.25× baseline	Fires March (RMSE $32.0k vs $24.4k baseline)
T3 — Systematic bias	|mean error| > 5% of median price	Fires Apr–Jul: bias –0.4% → **–7.9%** as the rebound accelerates
T4 — Interval health	80%-range coverage < 65%	Never fires, but coverage degrades 80% → ~73% — early warning visible

The verdict a real platform would act on: **retrain from March 2010** — the market recovered faster than the model's training window, and the model systematically prices below the market. Notably, drift detection alone (T1) shows *that* inputs changed; only the error and bias monitors (T2/T3) show it *matters* — the case for monitoring predictions, not just inputs.

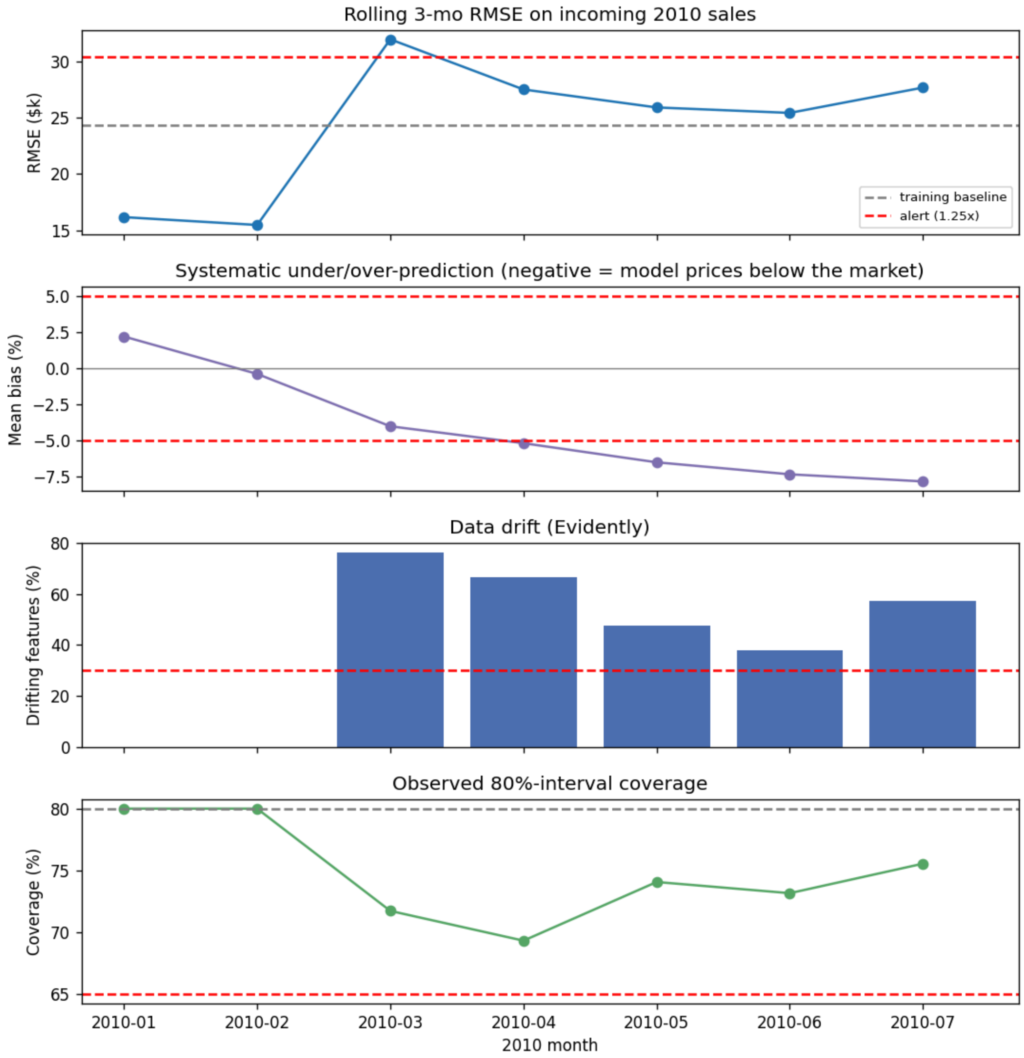

Figure 4 — Rolling RMSE, systematic bias, feature drift, and interval coverage over the 2010 stream; dashed red lines are the alert thresholds.

9. Pricing-strategy recommendations

1. **Price on quality and location first** — quality, area, and neighborhood (largely via school proximity) carry most of the signal; renovation history refines it.
2. **Show the range, not just the number.** A calibrated range converts the tool from a black box into a negotiation aid and tells agents when to escalate to an appraisal.
3. **Treat the model as perishable.** In a moving market it drifted beyond tolerance within ~3 months; budget for quarterly retraining, gated by the four triggers above.

10. Limitations and future work

- The contextual layer is synthetic; assumptions are documented and test-enforced, but real amenity and transaction data would change coefficients.
- Prices are expressed at the 2010 Ames market level; another market requires local data and recalibration.
- The cheapest price decile is over-predicted — a segment-specific model or monotonicity constraints are natural next steps.
- Retraining is recommended by triggers but not yet automated; the next step is a scheduled retrain–evaluate–promote loop.
- Monthly stream samples are small (6–48 sales), which bounds the confidence of monthly drift verdicts — hence rolling windows and minimum-sample gates.